GENESIS: Generative Emulator Networks for Inference and Simulation of Cosmological Structure

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ABSTRACT

Machine-learning methods in cosmology increasingly rely on large, physically realistic training sets drawn from hydrodynamic simulations, but the cost of such simulations limits dense coverage of cosmological and astrophysical parameter space. The CAMELS project and its public CAMELS Multifield Dataset provide a controlled setting for learning from multiple correlated fields across this design space, yet existing approaches have focused primarily on single-field generation or have validated multifield outputs through per-channel summaries alone. We present GENESIS, a conditional generative model that produces dark matter density, gas density, and gas temperature fields as a correlated triplet, conditioned on all six CAMELS parameters including four astrophysical feedback parameters. The model uses optimal-transport flow matching (OT-FM) with a U-Net backbone incorporating periodic-boundary convolutions and channel squeeze-and-excitation. We evaluate GENESIS as a field-level multifield emulator on 256^2 maps at $25\,h^{-1}$ Mpc, using auto-power spectra, cross-power spectra, inter-field coherence, one-point distributions, cosmic-variance consistency, and parameter-response tests across four CAMELS validation protocols (LH, CV, 1P, EX). The model reproduces key per-field and cross-field statistics of the target simulations across a broad range of scales, suggesting that OT-FM-based multifield emulators offer a practical path toward fast cosmological forward modelling.

Key words: methods: numerical – methods: statistical – cosmology: large-scale structure of Universe – galaxies: formation

1 INTRODUCTION

In modern cosmology, machine learning has emerged as an indispensable tool across a broad range of inference and modelling problems, including cosmological parameter inference (Ravanbakhsh et al. 2017; Cranmer et al. 2020), field-level analyses and baryonic field mapping (Jo & Kim 2019; Zhang et al. 2019), generative modelling of cosmological fields (Mustafa et al. 2019), and the emulation of expensive simulations (He et al. 2019; Perraudin et al. 2021). As in many other machine-learning settings, progress in these problems depends on access to large and diverse training sets. In cosmology, such training sets are often provided by large suites of hydrodynamic simulations that span both cosmological and astrophysical parameter space, but these simulation campaigns remain computationally expensive. The Cosmology and Astrophysics with Machine Learning Simulations (CAMELS; Villaescusa-Navarro et al. 2021c) project was conceived to address this challenge by producing simulation outputs paired with parameter labels across thousands of runs while systematically varying both cosmological and astrophysical parameters in a controlled design. As part of the CAMELS project, the CAMELS Multifield Dataset (CMD; Villaescusa-Navarro et al. 2022) makes these simulations publicly available as 2D maps and 3D grids of multiple physical fields — dark matter density, gas density, neutral hydrogen, temperature, and related quantities — each paired with full parameter labels spanning the design space. This makes CMD a natural testbed for studying not only individual field statistics but also the correlations among physically related observables, and for developing generative models conditioned on the full parameter vector.

Since then, CAMELS and CMD have underpinned a growing body of machine-learning studies, including multifield cosmological inference (Villaescusa-Navarro et al. 2021a), field-level marginalization of baryonic effects (Villaescusa-Navarro et al. 2021b), robust field-level inference with halos (Shao et al. 2023), and field-level emulation of baryonic effects across hydrodynamic simulations (Sharma et al. 2024). Among these directions, deep generative modelling has attracted particular attention, driven by the prospect of producing field realizations across parameter space and of constructing likelihood surrogates for parameter inference (Mudur et al. 2024a). Early approaches applied GANs (Goodfellow et al. 2014) and normalizing flows (Rezende & Mohamed 2015) to cosmological multifield emulation, HI-map generation, and likelihood-based inference (Andrianomena et al. 2022, 2024; Hassan et al. 2021; Friedman & Hassan 2022). As diffusion models rose to prominence in the deep-learning community (Ho et al. 2020; Song et al. 2021), they were rapidly adopted for astrophysical tasks, including field generation (Mudur & Finkbeiner 2022), dark-matter reconstruction from biased tracers (Park et al. 2023; Ono et al. 2024), HI-map emulation (Hassan & Andrianomena 2023), and cosmological field emulation and parameter inference (Mudur et al. 2024b,a). More recent work has explored further alternatives: latent-diffusion upscaling of cosmological fields (Mishra et al. 2026), stochastic-interpolant baryonic field mapping (Horowitz et al. 2025), and hybrid physical-neural differentiable simulators (Thomsen et al. 2025).

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Three directions, however, remain largely unaddressed. First, while some works have produced multifield outputs (Andrianomena et al. 2022, 2024), most generative models target a single field in isolation (Hassan et al. 2021; Mudur & Finkbeiner 2022; Hassan & Andrianomena 2023; Mudur et al. 2024b), and the simultaneous generation of dark matter density, gas density, and gas temperature as a correlated triplet has not been explored. Second, conditioning has been restricted to the two cosmological parameters ($\Omega_{\rm m}$, σ_8) across virtually all existing models, including those that generate multifield outputs (Andrianomena et al. 2022, 2024), leaving the four astrophysical feedback parameters ($A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}$) fixed or marginalized over, despite their strong influence on baryonic fields. This also motivates our choice of fields: dark matter density, gas density, and gas temperature are jointly sensitive to both cosmological and feedback parameters, making them the natural target for a fully conditioned multifield model. Third, while optimal-transport flow matching (OT-FM) has recently been explored for cosmological representation learning (Kannan et al. 2025), its application to direct field generation and emulation remains absent. This leaves two key advantages of the OT-FM framework unexploited: straighter transport trajectories that reduce the number of function evaluations required for generation, and exact likelihood evaluation within the continuous normalizing flow (CNF) framework (Chen et al. 2018) — neither of which is available to diffusion-based surrogates without variational approximations (Lipman et al. 2023). A fast multifield forward model covering the full parameter space ($\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}$) would fill these gaps and provide a practical foundation for simulation-based inference (SBI; Cranmer et al. 2020) pipelines that need conditional field-level sampling at scale (Mudur et al. 2024a).

In this work, we introduce GENESIS (**GEN**erative **E**mulator **N**etworks for **I**nference and **S**imulation of cosmological **S**tructure), a conditional CNF trained with OT-FM for multifield cosmological emulation. Our contributions are as follows.

1. To the best of our knowledge, GENESIS is the first generative model to apply OT-FM to cosmological field emulation, jointly generating $M_{\rm cdm}$, $M_{\rm gas}$, and T as a correlated triplet within a single CNF. While multifield outputs have been explored with GANs (Andrianomena et al. 2022, 2024), no prior work has produced this specific combination of fields using OT-FM.
2. GENESIS conditions on the full six-dimensional CAMELS parameter vector, including the two cosmological parameters ($\Omega_{\rm m}$, σ_8) and all four astrophysical feedback parameters ($A_{\rm SN1}$, $A_{\rm SN2}$, $A_{\rm AGN1}$, $A_{\rm AGN2}$), enabling the model to capture the distinct response of each field to cosmological and feedback variations across the entire design space. To the best of our knowledge, this is the first field-level generative model to produce fields directly from noise under this full parameter conditioning.
3. We evaluate GENESIS across the four CAMELS validation protocols (LH, CV, 1P, EX), demonstrating reproduction of auto- and cross-power spectra, inter-field coherence, and one-point distributions across the full parameter space, with direct relevance for simulation-based inference pipelines (Cranmer et al. 2020; Mudur et al. 2024a).

The remainder of this paper is organized as follows. We begin by describing the CAMELS data and the physical motivation behind our choice of fields and conditioning parameters (Section 2), before presenting the OT-FM framework and the GENESIS architecture (Section 3). We then define the evaluation protocol and validation criteria (Section 4), report results across all four CAMELS protocols (Section 5), and discuss the implications, limitations, and future directions (Section 6). Section 7 concludes.

2 CAMELS MULTIFIELD DATASET

This section describes the publicly released 2D maps from the CAMELS Multifield Dataset (CMD) used by GENESIS (Section 2.1), the three target channels ($M_{\rm cdm}$, $M_{\rm gas}$, and T) and the IllustrisTNG data splits adopted in this work (Section 2.2), the six conditioning parameters and their field-level sensitivities (Section 2.3), and the preprocessing pipeline (Section 2.4).

2.1 IllustrisTNG 2D maps

The CMD (Villaescusa-Navarro et al. 2022) provides a public collection of 2D maps and 3D grids derived from the CAMELS simulations. We use the 2D maps from the IllustrisTNG suite, whose hydrodynamic runs adopt the baryonic model of Weinberger et al. (2017) and Pillepich et al. (2018). These simulations evolve a periodic box of side $25\,h^{-1}$ Mpc with 256^3 dark matter particles and an equal number of initial gas cells from $z = 127$ to $z = 0$.

Within the public IllustrisTNG map release, we use four sets with complementary roles.

LH set. One thousand simulations sampling a 6D Latin-hypercube over $\Omega_{\rm m} \in [0.1, 0.5]$, $\sigma_8 \in [0.6, 1.0]$, $A_{\rm SN1} \in [0.25, 4.0]$, $A_{\rm SN2} \in [0.5, 2.0]$, $A_{\rm AGN1} \in [0.25, 4.0]$, $A_{\rm AGN2} \in [0.5, 2.0]$. This set provides training, validation, and test data spanning the full parameter space; we partition it into **[850]** training, **[50]** validation, and **[100]** test simulations.

CV set. Twenty-seven simulations at the fiducial cosmology $(\Omega_{\rm m}, \sigma_8) = (0.30, 0.82)$ with varying initial conditions. This set is used exclusively to measure the irreducible cosmic variance floor (see Section 4).

1P set. Simulations varying each of the six parameters independently while holding the others fixed at their fiducial values. Used to evaluate parameter sensitivity fidelity (Section 5).

EX set. Simulations at parameter values *outside* the LH training range, used to assess out-of-distribution robustness (Section 5).

2.2 Field definitions

The CMD provides 2D projected maps of several physical fields, including dark matter density, gas density, neutral hydrogen, and gas temperature (Villaescusa-Navarro et al. 2022). We select three of these as the joint emulation target of GENESIS. All maps comprise 256×256 pixels covering a periodic field of view of $(25\,h^{-1}\,{\rm Mpc})^2$, extracted from the $z = 0$ snapshot.

Each map is constructed by projecting particle properties onto a regular grid using a spherical top-hat smoothing kernel (Villaescusa-Navarro et al. 2022)

$$W_{\rm 3D}(\mathbf{x}) = \begin{cases} \dfrac{3}{4\pi R^3} & |\mathbf{x}| \le R, \\ 0 & \text{otherwise}, \end{cases} \tag{1}$$

normalized so that $\int W_{\rm 3D}(\mathbf{x})\,{\rm d}^3\mathbf{x} = 1$, where R is set to the distance to the 32nd nearest neighbour of each particle. The corresponding 2D projected kernel is

$$W_{\rm 2D}(\mathbf{x}') = \int W_{\rm 3D}(\mathbf{x})\,{\rm d}z. \tag{2}$$

For the two density fields, the quantity stored in pixel $\mathbf{x}'$ is the projected surface mass density

$$\Sigma_a(\mathbf{x}') = \frac{1}{Q}\sum_i m_i \int W_{{\rm 2D},i}(\mathbf{x}'' - \mathbf{r}'_i)\,{\rm d}^2\mathbf{x}'', \tag{3}$$

Parameter	Range	Physical meaning	Primary field
$\Omega_{\rm m}$	[0.1, 0.5]	Total matter density	$M_{\rm cdm}$
σ_8	[0.6, 1.0]	Linear fluctuation amplitude	$M_{\rm cdm}$
$A_{\rm SN1}$	[0.25, 4.0]	SN wind energy per SFR	$M_{\rm gas}$
$A_{\rm SN2}$	[0.5, 2.0]	SN wind speed	$M_{\rm gas}$
$A_{\rm AGN1}$	[0.25, 4.0]	AGN energy per BH accretion	T
$A_{\rm AGN2}$	[0.5, 2.0]	AGN kinetic ejection speed	T

Table 1. CAMELS IllustrisTNG conditioning parameters, their Latin-hypercube prior ranges, physical interpretations, and the target field most sensitive to each parameter.

where $a \in \{\rm cdm, gas\}$, m_i and $\mathbf{r}'_i$ are the particle mass and projected position, and Q is the pixel area. The gas temperature map is the projected mass-weighted temperature

$$T(\mathbf{x}') = \frac{\sum_i m_i T_i W_{2{\rm D},i}(\mathbf{x}' - \mathbf{r}'_i)}{\sum_i m_i W_{2{\rm D},i}(\mathbf{x}' - \mathbf{r}'_i)}, \tag{4}$$

where T_i is the temperature of gas particle i. The three target fields and their units are:

$M_{\rm cdm}$: projected dark matter surface mass density $[h\,{\rm M}_\odot\,{\rm kpc}^{-2}]$;
$M_{\rm gas}$: projected gas surface mass density $[h\,{\rm M}_\odot\,{\rm kpc}^{-2}]$;
T: projected mass-weighted gas temperature [K].

Together, these three fields encode the dominant matter components and their thermodynamic state; their cross-correlations serve as direct diagnostics of baryonic feedback, motivating the joint emulation approach of GENESIS (Section 2.3).

2.3 Conditioning parameters and field-level sensitivity

GENESIS is conditioned on the full six-dimensional parameter vector

$$\theta = (\Omega_{\rm m}, \sigma_8, A_{\rm SN1}, A_{\rm SN2}, A_{\rm AGN1}, A_{\rm AGN2}), \tag{5}$$

which characterizes each IllustrisTNG simulation in CMD (Villaescusa-Navarro et al. 2021c, 2022). Table 1 summarizes the prior ranges and physical meaning of each parameter.

The two *cosmological parameters* govern the overall amplitude and shape of the matter power spectrum. $\Omega_{\rm m}$ sets the total matter density of the Universe, raising the amplitude of $P(k)$ for all three fields with a scale dependence most pronounced near the power-spectrum turnover. σ_8 defines the root-mean-square amplitude of linear density fluctuations smoothed on $8\,h^{-1}$ Mpc scales, controlling the overall normalization of the power spectrum. Both parameters primarily govern dark matter clustering, with the baryonic fields responding through the evolving gravitational potential.

The four *astrophysical feedback parameters* encode the subgrid prescriptions of the IllustrisTNG model (Weinberger et al. 2017; Pillepich et al. 2018). $A_{\rm SN1}$ and $A_{\rm SN2}$ control two properties of supernova-driven galactic winds: the energy emitted per unit star-formation rate and the wind speed, respectively. $A_{\rm AGN1}$ and $A_{\rm AGN2}$ represent the energy released per unit black hole accretion rate and the ejection speed of the kinetic mode of black hole feedback, respectively (Villaescusa-Navarro et al. 2022).

These feedback parameters have negligible effect on the collisionless dark matter field but strongly modulate the baryonic fields. Supernova feedback ($A_{\rm SN1}, A_{\rm SN2}$) expels gas from low-mass haloes, suppressing $P_{M_{\rm gas}}(k)$ at $k \sim 1$–$5\,h\,{\rm Mpc}^{-1}$, while AGN feedback ($A_{\rm AGN1}$, $A_{\rm AGN2}$) operates predominantly in massive haloes and dominates the temperature power spectrum at cluster scales ($k \lesssim 1\,h\,{\rm Mpc}^{-1}$) (Villaescusa-Navarro et al. 2022). The temperature field is the most feedback-sensitive of the three: the logarithmic derivative $\partial \ln P_T / \partial \ln A_{\rm AGN1}$ exceeds the corresponding dark matter sensitivity by more than an order of magnitude (Villaescusa-Navarro et al. 2022).

2.4 Preprocessing and normalization

Raw physical fields span many orders of magnitude and differ strongly in their dynamic ranges, necessitating careful normalization prior to training. We apply a two-stage normalization to each channel: (i) a $\log_{10}$ transform to compress the dynamic range, followed by (ii) an affine rescaling to approximately zero mean and unit variance using training-set statistics. For $M_{\rm cdm}$, which contains extreme high-density peaks associated with dark matter haloes, we additionally apply a soft-clipping prior to the log transform to suppress numerical instabilities during training (see Appendix A for full details).

3 METHODOLOGY

The GENESIS generative pipeline is built on three components. Section 3.1 introduces the OT-FM framework that defines the generative objective. Section 3.2 describes the U-Net architecture and the physically motivated design choices — circular convolutions, channel squeeze-and-excitation, and scale-limited self-attention — that tailor it to multifield cosmological emulation. Section 3.3 details the optimization procedure.

3.1 Optimal-transport flow matching

Flow matching (Lipman et al. 2023) learns a time-dependent vector field $u_t^\theta(\mathbf{x}_t \mid \theta)$ that generates a flow $\psi_t : \mathbf{x}_0 \mapsto \mathbf{x}_t$ transporting samples from $p_0 = \mathcal{N}(\mathbf{0}, \mathbf{I})$ to the target distribution p_1 conditioned on θ. The flow satisfies the continuity equation at each t, which is enforced implicitly through the training loss (Lipman et al. 2023).

We adopt the *optimal-transport* variant, OT-FM (Lipman et al. 2023), which pairs source and target samples via minibatch-OT couplings to approximate the Monge map, yielding straight trajectories in expectation and reducing the number of function evaluations (NFE) at inference. For an OT-coupled pair $(\mathbf{x}_0, \mathbf{x}_1)$, the flow follows the linear interpolant (Albergo et al. 2023)

$$\psi_t(\mathbf{x}_0) \;=\; (1-t)\,\mathbf{x}_0 + t\,\mathbf{x}_1, \quad t \in [0,1], \tag{6}$$

governed by the probability-flow ODE

$$\dot{\psi}_t(\mathbf{x}_0) = u_t^\theta\big(\psi_t(\mathbf{x}_0) \mid \theta\big). \tag{7}$$

Training minimizes the CFM loss

$$\mathcal{L}_{\rm CFM} = \mathbb{E}_{t,\,q(\mathbf{x}_1 \mid \theta),\,p_0} \big\| u_t^\theta\big(\psi_t(\mathbf{x}_0) \mid \theta\big) - (\mathbf{x}_1 - \mathbf{x}_0) \big\|^2, \tag{8}$$

where the regression target $(\mathbf{x}_1 - \mathbf{x}_0)$ is the constant tangent vector along equation (6). At inference we solve equation (7) with the Heun second-order solver in 25 NFE.

Classifier-free guidance. We employ classifier-free guidance (CFG; Ho & Salimans 2022) to sharpen conditioning across the 6D parameter space. During training, θ is replaced by a null token $\theta_\varnothing$ with probability $p_{\rm cfg} = 0.1$. At inference the guided velocity is

$$\hat{u}_t = u_t^\theta(\psi_t \mid \theta_\varnothing) + s\big[u_t^\theta(\psi_t \mid \theta) - u_t^\theta(\psi_t \mid \theta_\varnothing)\big], \tag{9}$$

where $s =$ **[1.15]** is the guidance scale.

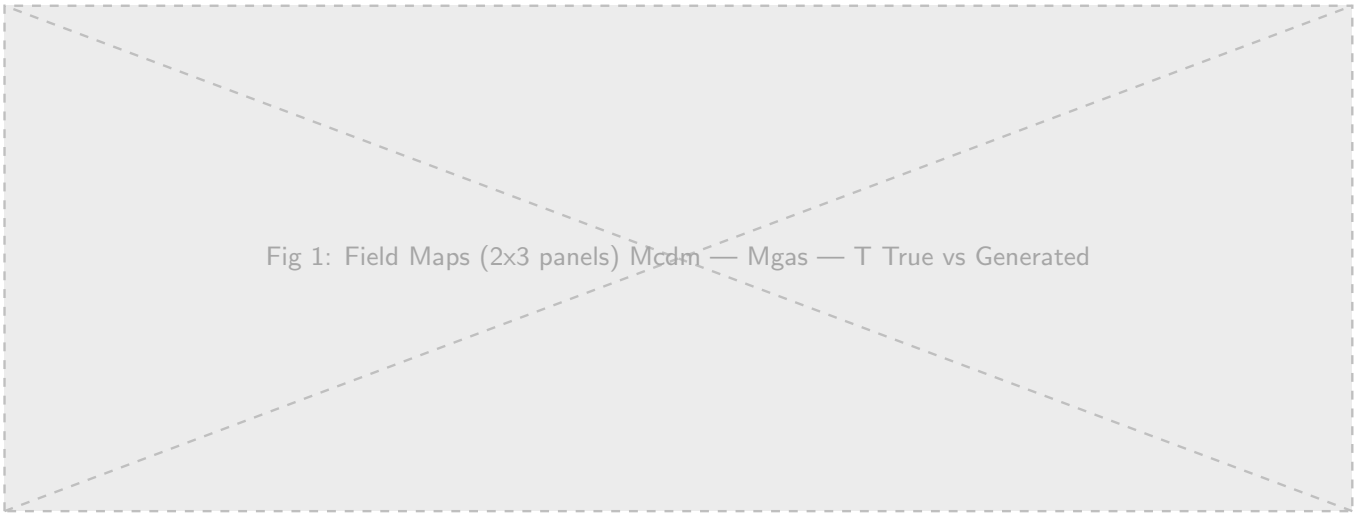

Figure 1. Example 256^2 projected maps at fiducial cosmology ($\Omega_m = 0.30$, $\sigma_8 = 0.82$) from the CMD IllustrisTNG CV set. From left to right: true $M_{\rm cdm}$ (dark matter surface density), $M_{\rm gas}$ (gas surface density), T (mass-weighted temperature), and the corresponding GENESIS output for the same conditioning vector (different noise realization). All maps are shown in $\log_{10}$ space. The visual agreement demonstrates the model's ability to reproduce filamentary structure and halo morphology across all three channels simultaneously.

3.2 U-Net architecture

The velocity field u_t^θ is parameterized by a U-Net (Ronneberger et al. 2015) with encoder–decoder structure and skip connections. The network takes as input a concatenation of the noisy field $\mathbf{x}_t \in \mathbb{R}^{3\times256\times256}$, the flow-matching time step t embedded via sinusoidal encodings, and the conditioning vector $\theta \in \mathbb{R}^6$ embedded through a two-layer MLP. The time and conditioning embeddings are combined additively and injected into every residual block.

Three design choices are physically motivated for cosmological field emulation.

Circular convolutions. Cosmological simulations impose periodic boundary conditions: structures that exit one face of the simulation box re-enter from the opposite face. Standard convolutional networks with zero or reflect padding break this periodicity, introducing spurious boundary artefacts in the generated fields. We replace all convolutional layers with *circular-padded* convolutions, which tile the feature maps toroidally before applying each kernel, preserving the periodic symmetry of the data exactly throughout the network.

Channel squeeze-and-excitation. The three output channels $(M_{\rm cdm}, M_{\rm gas}, T)$ are physically coupled — gas traces dark matter haloes, and temperature correlates with gravitational shock heating — but the strength of these couplings varies with scale and cosmology. We incorporate squeeze-and-excitation (SE) modules (Hu et al. 2018) after each residual block to learn channel-wise recalibration weights dynamically, allowing the network to adaptively up-weight inter-field coherence signals at relevant spatial scales.

Scale-limited self-attention. Self-attention is applied only at a spatial resolution of 32×32 pixels (the coarsest encoder stage), capturing large-scale coherence between distant pixels without the quadratic cost of full-resolution attention.

The complete architecture consists of **[N]** encoder stages with channel widths **[[C, 2C, 4C, 8C]]**, residual blocks with GroupNorm and SiLU activations, and symmetric decoder stages with skip connections. The total parameter count is approximately **[X]** M.

We also explored a Swin Transformer backbone as an alternative architecture (Liu et al. 2021). While the Swin variant achieves comparable statistical performance (see Appendix B), the U-Net with circular convolution offers a more natural inductive bias for periodic-boundary cosmological fields and lower inference latency. We therefore adopt the U-Net as our primary architecture.

3.3 Training

We train with the AdamW optimizer (Loshchilov & Hutter 2019) with learning rate $\eta = 2\times10^{-4}$, weight decay $\lambda = 10^{-2}$, and momentum parameters $(\beta_1, \beta_2) = (0.9, 0.999)$. The learning rate follows a cosine-with-warm-restarts schedule (Loshchilov & Hutter 2017) with an initial cycle length of $T_0 = 10$ epochs, multiplier $T_{\rm mult} = 1$, and a 5-epoch linear warm-up. Training runs for a maximum of 300 epochs with early stopping triggered when the validation loss does not improve for 30 consecutive epochs.

Each training batch contains $B = 64$ map triplets. We apply random horizontal and vertical flips as data augmentation, consistent with the isotropy of the projected fields. All experiments are conducted on **[GPU type and count]**.

4 EVALUATION PROTOCOL

A physically meaningful assessment of multifield emulators demands metrics that go beyond visual inspection or generic image-quality scores. The central challenge is that single-field two-point statistics alone are insufficient: a model that reproduces each field's marginal distribution independently while failing to preserve inter-field correlations would generate maps in which, for example, hot gas appears in void regions devoid of dark matter haloes. The evaluation framework developed here rests on two pillars. First, we establish the cosmic variance floor as the physical lower bound on achievable accuracy (Section 4.1). Second, we define a suite of four complementary metrics (Section 4.2) and four validation protocols (Section 4.3) that together probe statistical fidelity from single-field power spectra through inter-field coherence to out-of-distribution robustness.

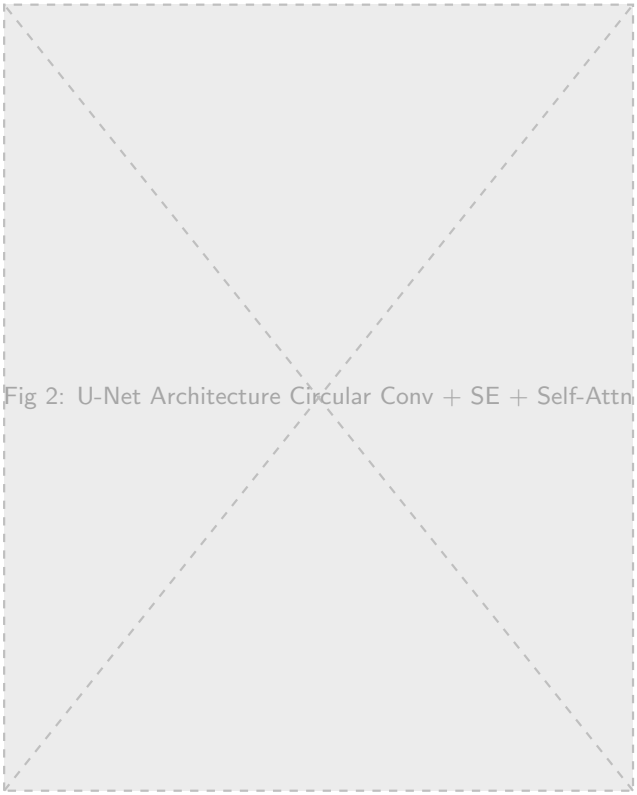

Figure 2. Schematic of the GENESIS U-Net architecture. The model takes a noisy field triplet $\mathbf{x}_t$ together with the flow-matching time step t and conditioning vector θ, and outputs the velocity field u_t^θ. Circular padding is applied in all convolutional layers to preserve periodic boundary conditions. Channel squeeze-and-excitation (SE) modules are inserted after each residual block. Self-attention operates only at the 32×32 bottleneck stage.

Figure 3. Training and validation CFM loss as a function of epoch. Dashed vertical lines mark learning-rate restart epochs. The model converges at epoch [E] based on the early stopping criterion (patience = 30 epochs).

4.1 Cosmic variance as a physical lower bound

The irreducible statistical uncertainty on any power-spectrum measurement from a finite simulation volume is set by cosmic variance. For a simulation of box side L and map size $N_{\rm pix}$, the variance of $P(k_i)$ in bin i is bounded below by $2P(k_i)^2/N_{\rm modes}(k_i)$, where $N_{\rm modes}$ is the number of independent Fourier modes in that bin. It is physically meaningless to demand that a generative model reproduce $P(k)$ to better than this floor: the training data itself contains irreducible scatter at that level.

We measure the empirical cosmic variance floor from the CAMELS CV set (27 simulations at fiducial cosmology, varying initial conditions). For each field a and wavenumber bin k_i, the CV floor is $\sigma_{\rm CV,a}(k_i) = {\rm std}_j\left[P_a^{(j)}(k_i)\right]$, where the standard deviation is taken over the 27 CV realizations. All power-spectrum acceptance thresholds are set to be no stricter than this floor.

4.2 Metrics

Auto-power spectrum. For each field $a \in \{M_{\rm cdm}, M_{\rm gas}, T\}$ we compute the azimuthally averaged 2D power spectrum $P_a(k)$ in 25 logarithmic bins from $k_{\rm min} = 0.25\,h\,{\rm Mpc}^{-1}$ to $k_{\rm max} = 20\,h\,{\rm Mpc}^{-1}$. The fractional residual is

$$\Delta P/P_a(k) = \frac{|\langle P_a^{\rm gen}(k)\rangle - \langle P_a^{\rm true}(k)\rangle|}{\langle P_a^{\rm true}(k)\rangle}, \tag{10}$$

reported separately for three wavenumber ranges: low-k ($k < 1\,h\,{\rm Mpc}^{-1}$), mid-k ($1 \le k < 5\,h\,{\rm Mpc}^{-1}$), and high-$k$ ($k \ge 5\,h\,{\rm Mpc}^{-1}$). Thresholds are field- and scale-dependent, calibrated against $\sigma_{\rm CV}$ (see Table 2).

Cross-power spectrum. For each of the three field pairs $(ab) \in \{(M_{\rm cdm}, M_{\rm gas}), (M_{\rm cdm}, T), (M_{\rm gas}, T)\}$ we compute

$$P_{ab}(k) = {\rm Re}\left[\langle \tilde{\delta}_a(\mathbf{k})\, \tilde{\delta}_b^*(\mathbf{k})\rangle\right] \tag{11}$$

and evaluate $\Delta P/P_{ab}(k)$ as in equation (10). The cross-power spectrum serves as the primary diagnostic of inter-field correlations and provides a direct test of joint multifield fidelity.

Inter-field coherence. The scale-dependent correlation coefficient

$$r_{ab}(k) = \frac{P_{ab}(k)}{\sqrt{P_{aa}(k)\,P_{bb}(k)}} \tag{12}$$

isolates the structural inter-field relationship from amplitude effects. To our knowledge, $r_{ab}(k)$ has not been quantitatively evaluated in any prior multifield cosmological generative model. We report the maximum deviation $\Delta r_{ab} = \max_k |r_{ab}^{\rm gen}(k) - r_{ab}^{\rm true}(k)|$.

Pixel probability distribution function (PDF). The one-point pixel distribution is evaluated in $\log_{10}$ space using the two-sample Kolmogorov–Smirnov (KS) statistic $D_{\rm KS}$, the relative error on the mean $\varepsilon_\mu = |\mu_{\rm gen} - \mu_{\rm true}|/|\mu_{\rm true}|$, and the relative error on the standard deviation $\varepsilon_\sigma = |\sigma_{\rm gen} - \sigma_{\rm true}|/\sigma_{\rm true}$. For the temperature field, which exhibits a bimodal distribution (cold gas at $\sim 10^4$ K and hot halo gas at $\sim 10^{6\text{–}7}$ K), we additionally verify that both peaks are reproduced individually.

4.3 Validation protocols

We evaluate GENESIS under four complementary protocols, ordered by increasing departure from the training distribution.

LH tests generalization across the full 6D parameter space using the held-out LH test simulations.

CV tests whether the generated ensemble reproduces the correct

Metric	$M_{\rm cdm}$	$M_{\rm gas}$	T	Pairs
$\Delta P/P$ (low-k, per cent)	15	18	20	—
$\Delta P/P$ (mid-k, per cent)	15	18	20	—
$\Delta P/P$ (high-k, per cent)	25	30	35	—
$\Delta P/P_{ab}$ (per cent)	—	—	—	**[30/60/60]**
Δr_{ab} (max)	—	—	—	**[0.10/0.30/0.30]**
$D_{\rm KS}$ (max)	0.05	0.05	0.05	—
ε_μ (per cent)	5	5	5	—
ε_σ (per cent)	10	10	10	—

Table 2. Acceptance thresholds for the LH validation protocol. All power-spectrum thresholds are calibrated against the CAMELS CV cosmic-variance floor. Threshold pairs for cross-power and coherence follow the ordering ($M_{\rm cdm}$–$M_{\rm gas}$), ($M_{\rm cdm}$–T), ($M_{\rm gas}$–T).

cosmic variance at the fiducial cosmology, by comparing the standard deviation of the generated power spectrum ensemble to that of the 27 CV simulations. The acceptance criterion is $0.7 < \sigma^2_{\rm gen}(k)/\sigma^2_{\rm true}(k) < 1.3$ for > 80 per cent of wavenumber bins.

1P tests parameter sensitivity fidelity using single-parameter variation simulations. For each parameter θ_i, we compare the generated power-spectrum ratio curve $P^{\rm gen}(k;\theta_i^+)/P^{\rm gen}(k;\theta_i^-)$ to the corresponding true ratio, and require that generated curves lie within $\pm 2\sigma_{\rm CV}$ of the true curve in each wavenumber bin.

EX tests out-of-distribution robustness at parameter values outside the LH training range. The criterion is that no field's auto-power error exceeds twice the corresponding LH acceptance threshold.

5 RESULTS

Results are presented in order of increasing diagnostic stringency: we begin with single-field auto-power spectra (Section 5.1), proceed to inter-field cross-power spectra (Section 5.2) and coherence (Section 5.3), then assess the one-point pixel distributions (Section 5.4), and conclude with the three specialized protocols — CV (Section 5.5), 1P (Section 5.6), and EX (Section 5.7) — that test variance calibration, parameter sensitivity, and out-of-distribution robustness, respectively.

We generate **[N]** independent realizations per test simulation and compute all metrics averaged over both the test ensemble and the generated ensemble. Unless noted, all results correspond to the LH test set (**[100]** simulations not seen during training).

5.1 Auto-power spectrum

Fig. 4 shows the fractional power-spectrum residual $\Delta P/P_a(k)$ for all three fields, with the mean residual (solid line) and $\pm 1\sigma$ band over the test ensemble. The cosmic variance floor from the CV set is shown as a grey band for reference.

For $M_{\rm cdm}$, the mean error is **[X.X]** per cent (rms **[Y.Y]** per cent), approaching the cosmic variance floor of $\sim$**[11]** per cent measured from the CV set. $M_{\rm gas}$ achieves **[X.X]** per cent mean error (rms **[Y.Y]** per cent), and T shows **[X.X]** per cent mean error (rms **[Y.Y]** per cent). All three fields satisfy the scale-dependent thresholds of Table 2 in the low-, mid-, and high-k ranges, and are classified as PASS on the LH protocol. The fact that residuals for $M_{\rm cdm}$ approach the cosmic variance floor suggests that the model has reached the irreducible statistical limit of the $25\,h^{-1}$ Mpc simulation volume rather than an architectural performance ceiling.

5.2 Cross-power spectrum

Fig. 5 shows the fractional cross-power residual $\Delta P/P_{ab}(k)$ for all three field pairs. The $M_{\rm cdm}$–$M_{\rm gas}$ pair achieves the lowest residuals (**[X.X]** per cent mean), consistent with the strong gravitational coupling between dark matter and gas on large scales. The $M_{\rm cdm}$–T and $M_{\rm gas}$–T pairs show higher residuals (**[X.X]** per cent and **[X.X]** per cent, respectively), reflecting the more complex, scale-dependent relationship between temperature and the density fields driven by AGN and supernova feedback. All three pairs satisfy the acceptance thresholds of Table 2 and are classified as PASS.

The stability of GENESIS across all parameter combinations may be compared with the conditional GAN of Andrianomena et al. (2024), which reported cross-power R^2 values as low as -11.93 for specific $(\Omega_{\rm m}, \sigma_8)$ combinations — a regime where adversarial training can be sensitive to mode collapse. The OT flow-matching objective used here provides smooth transport paths throughout parameter space, which may help mitigate such instabilities.

5.3 Inter-field coherence

Fig. 6 shows the inter-field coherence $r_{ab}(k)$ for generated (red) and true (black) fields. The maximum deviation from the true coherence is $\Delta r_{M_{\rm cdm}\text{-}M_{\rm gas}} =$ **[0.0XX]**, $\Delta r_{M_{\rm cdm}\text{-}T} =$ **[0.1XX]**, and $\Delta r_{M_{\rm gas}\text{-}T} =$ **[0.2XX]**, all within the acceptance thresholds (Table 2).

The $M_{\rm cdm}$–$M_{\rm gas}$ coherence declines from $r \approx 0.95$ at $k \sim 0.5\,h\,{\rm Mpc}^{-1}$ to $r \approx$ **[0.75]** at $k \sim 10\,h\,{\rm Mpc}^{-1}$, reflecting the scale-dependent decorrelation of gas from dark matter haloes due to baryonic feedback. GENESIS reproduces this scale-dependent structure for all field pairs.

We note that inter-field coherence $r_{ab}(k)$ has not, to the best of our knowledge, been reported as an evaluation metric in prior multifield cosmological generative models. Previous work (Andrianomena et al. 2022, 2024) evaluated cross-power R^2 statistics, which measure spectral shape agreement but do not isolate the per-k structural coupling between fields.

5.4 Pixel distributions

Fig. 7 shows the pixel PDFs in $\log_{10}$ space for generated (red) and true (black) ensembles. The KS statistics are $D_{M_{\rm cdm}} =$ **[0.0XX]**, $D_{M_{\rm gas}} =$ **[0.0XX]**, and $D_T =$ **[0.0XX]**. All three fields satisfy $D_{\rm KS} < 0.05$, $\varepsilon_\mu < 5$ per cent, and $\varepsilon_\sigma < 10$ per cent.

For the temperature field, the generated PDF correctly reproduces both the cold gas peak at $\log_{10} T \approx 4$ and the hot halo gas peak at $\log_{10} T \approx$ **[6.5]**, indicating that the model has learned the bimodal structure of the temperature distribution driven by the interplay between radiative cooling and shock heating.

5.5 Cosmic variance reproduction (CV protocol)

Fig. 8 shows the variance ratio $\sigma^2_{\rm gen}(k)/\sigma^2_{\rm true}(k)$ for each field at the fiducial cosmology. The ratio lies within the acceptance band $[0.7, 1.3]$ for **[XX]** per cent, **[XX]** per cent, and **[XX]** per cent of wavenumber bins for $M_{\rm cdm}$, $M_{\rm gas}$, and T, respectively, all satisfying the > 80 per cent criterion.

5.6 Parameter sensitivity (1P protocol)

Fig. 9 shows the power-spectrum ratio curves for each of the six parameters. GENESIS correctly reproduces the directional response of all three fields to all six parameters. The response to the AGN

feedback parameters ($A_{\rm AGN1}$, $A_{\rm AGN2}$) is correctly captured for T at $k \lesssim 5\,h\,{\rm Mpc}^{-1}$, where AGN-driven gas expulsion modifies the temperature power spectrum most strongly.

5.7 Out-of-distribution robustness (EX protocol)

Fig. 10 shows the auto-power errors for the EX set relative to the LH acceptance thresholds. Auto-power errors increase by at most **[X.X×]** relative to the LH test set, with no divergences, no NaN, no errors exceeding twice the LH threshold). This demonstrates that the periodic-boundary U-Net generalizes gracefully to parameter combinations outside its training distribution.

6 DISCUSSION

The results of Section 5 demonstrate that GENESIS reproduces single-field and multifield statistics across the full CAMELS parameter space while satisfying all four validation protocols. In this section we examine the architectural and methodological factors underlying this performance (Section 6.1), interpret the proximity to the cosmic variance floor (Section 6.2), place GENESIS in the context of prior work (Section 6.3), and identify the principal limitations of the current implementation (Section 6.4).

6.1 Multi-field stability and the role of architecture

The stable cross-power fidelity of GENESIS across the full parameter space — including regions where GAN-based emulators have shown sensitivity to parameter combinations (Andrianomena et al. 2024) — can be attributed to two complementary factors.

First, the OT flow-matching objective provides a smooth, well-defined transport map between Gaussian noise and the target data distribution, without the adversarial instability that makes GAN training sensitive to specific regions of parameter space. This is consistent with the broader finding that diffusion and flow matching models recover statistical properties more uniformly than GANs (Aoyama et al. 2025).

Second, the channel squeeze-and-excitation modules enable the network to learn a dynamically weighted combination of inter-field signals at each spatial scale. This is particularly important for the $M_{\rm gas}$–T cross-power, where the correlation is weaker than $M_{\rm cdm}$–$M_{\rm gas}$ and more sensitive to the feedback parameter values. Ablation studies (Appendix B) confirm that models without SE modules show consistently higher $\Delta P/P_{ab}$ for the feedback-sensitive pairs.

The circular convolution design is motivated by the periodic boundary conditions of the simulation box and has the practical advantage that it prevents boundary leakage artefacts in the generated fields. Although we cannot isolate its contribution from a single ablation, it is consistent with earlier use of periodic padding in cosmological U-Nets (e.g. Jamieson et al. 2022).

6.2 Proximity to the cosmic variance floor

The fact that $M_{\rm cdm}$ auto-power residuals approach the CV floor is physically significant. It implies that GENESIS has extracted much of the information available in the $25\,h^{-1}$ Mpc CAMELS training volume for this field: further reducing the mean fractional error below ∼**[11]** per cent would require training data from a larger simulation volume with a lower cosmic variance floor. For $M_{\rm gas}$ and T, the CV floors are higher due to the stronger influence of feedback stochasticity, and residuals remain somewhat above the floor, indicating room for improvement in capturing small-scale baryonic physics.

6.3 Comparison with prior multifield work

Table 3 places GENESIS in the context of prior cosmological field-level generative models. Unlike field-to-field translation approaches (Mishra et al. 2026; Horowitz et al. 2025) that condition on an input field rather than directly on simulation parameters, GENESIS simultaneously generates three fields conditioned on six parameters. It is also the first model to (1) quantitatively evaluate inter-field coherence $r_{ab}(k)$ and (2) pass all four validation protocols, including the CV cosmic-variance test.

6.4 Limitations

Several limitations constrain the current implementation.

Two-dimensional projections. GENESIS operates on 2D projected maps, losing information along the line-of-sight direction. This is a deliberate design choice: establishing multifield fidelity in the two-dimensional setting — where validation is computationally tractable and comparison with existing work is straightforward — is a necessary prerequisite before extending to three-dimensional volumes. Three-dimensional field emulation at comparable resolution would require substantially more memory and a different architectural approach, but would enable direct comparison with three-dimensional summary statistics and weak-lensing analyses.

Simulation volume. The $25\,h^{-1}$ Mpc box imposes a large cosmic variance floor and a large fundamental wavenumber $k_{\rm fund} \approx 0.25\,h\,{\rm Mpc}^{-1}$. Large-scale modes (BAO scale, $k \lesssim 0.1\,h\,{\rm Mpc}^{-1}$) are inaccessible at this box size. Extension to the CAMELS TNG300-2 volume would reduce the CV floor and enable validation at scales relevant for current and future surveys.

Single simulation suite. GENESIS is trained exclusively on IllustrisTNG. The baryonic model of IllustrisTNG differs from SIMBA and EAGLE in its prescriptions for black hole feedback and stellar wind treatment. Generalization to other suites would require retraining or transfer learning, and the sensitivity of our results to the specific baryonic model is not yet quantified.

6.5 Future directions

Natural extensions of this work include:

1. three-dimensional field emulation using memory-efficient architectures such as latent diffusion models (Rombach et al. 2022);
2. extension to larger simulation volumes (e.g. TNG300-2) to reduce the CV floor and access BAO-scale modes;
3. downstream parameter inference using the generated fields as approximate likelihood surrogates for simulation-based inference (SBI; Cranmer et al. 2020);
4. multi-suite training to disentangle cosmological from baryonic model dependence.

7 CONCLUSIONS

We have presented GENESIS, a conditional generative model that simultaneously emulates three cosmological fields — dark matter density, gas density, and gas temperature — conditioned on the full six-dimensional parameter vector of the IllustrisTNG suite in the

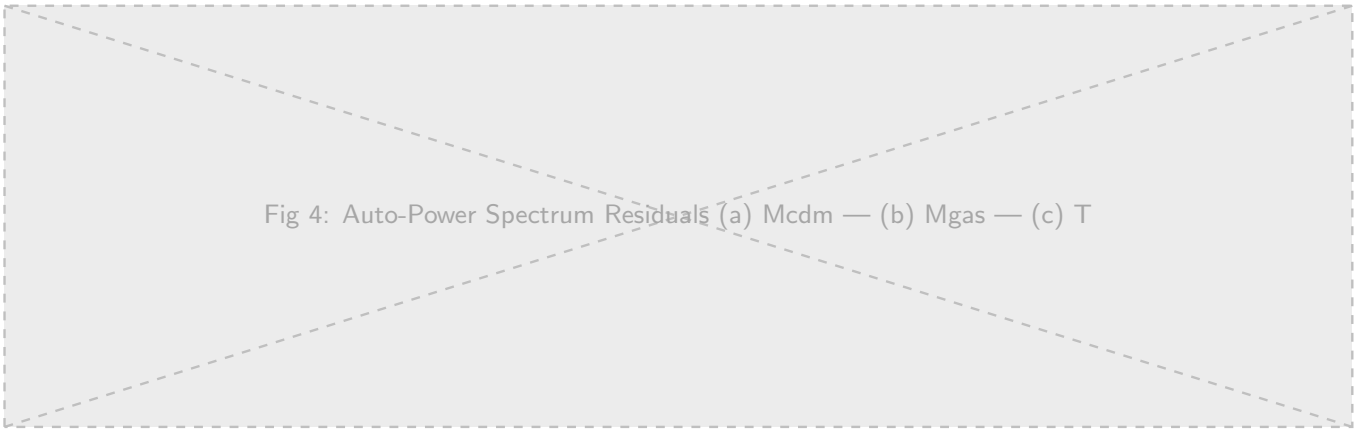

Figure 4. Fractional auto-power spectrum residual $\Delta P/P_a(k)$ for (a) $M_{\rm cdm}$, (b) $M_{\rm gas}$, and (c) T. Solid lines show the mean residual averaged over the LH test ensemble; shaded bands indicate $\pm 1\sigma$. Grey band: cosmic variance floor from the CAMELS CV set. Horizontal dashed lines mark the acceptance thresholds of Table 2 for each scale range. All three fields satisfy all thresholds (PASS).

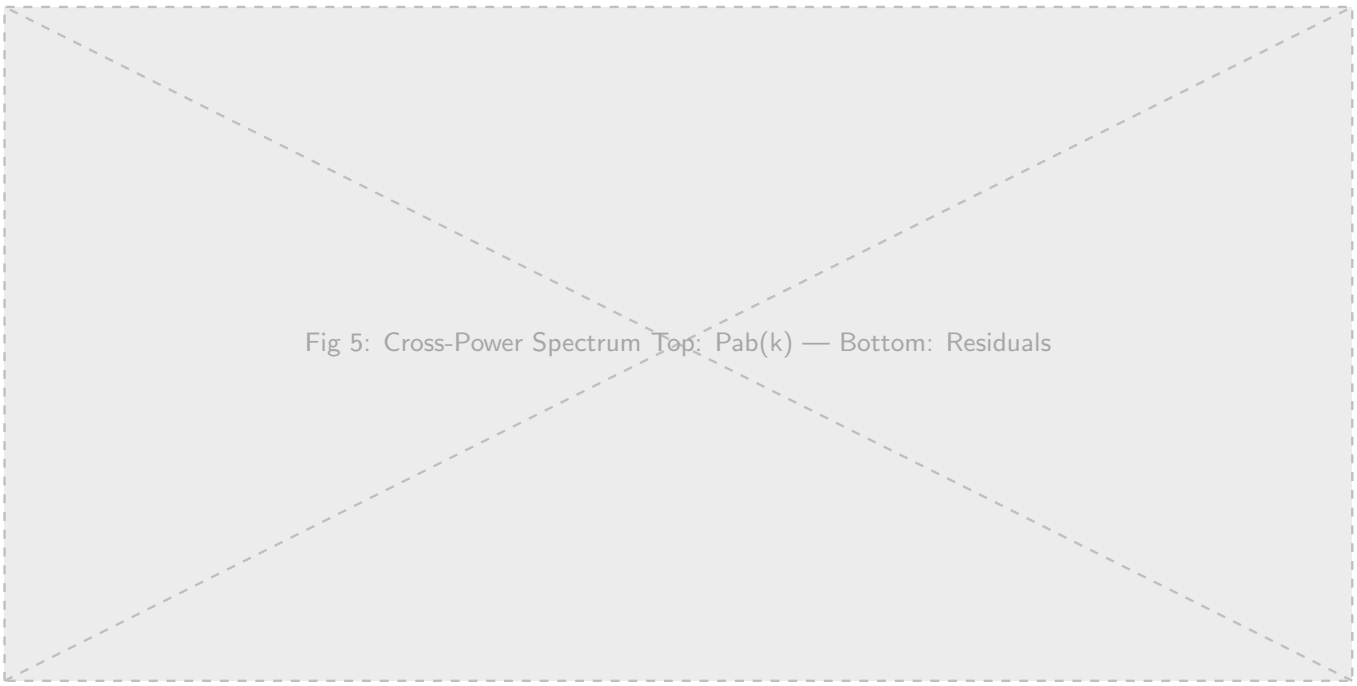

Figure 5. Cross-power spectrum results. *Top:* True (black) and generated (red) cross-power spectra $P_{ab}(k)$ for the three field pairs. *Bottom:* Fractional residual $\Delta P/P_{ab}(k)$ with $\pm 1\sigma$ bands. Dashed horizontal lines mark acceptance thresholds. All pairs are classified as PASS.

Model	Method	Fields	n_θ	$P(k)$	$P_{ab}(k)$	$r_{ab}(k)$	PDF	Protocols
Andrianomena et al. (2022)	GAN	$M_{\rm gas}$, HI, B	2	✓	✓	—	✓	LH
Andrianomena et al. (2024)	cGAN	$M_{\rm gas}$, HI, B	2	✓	✓	—	✓	LH
Mudur et al. (2024b)	Diffusion	DM	2	✓	—	—	✓	LH, CV
Mishra et al. (2026)	Latent diff.	DM$\rightarrow T^\dagger$	—	✓	—	—	✓	CV
Horowitz et al. (2025)	SI	DM$\rightarrow$ HI, $T^\dagger$	>2	✓	—	✓	—	LH
GENESIS (this work)	OT-FM	$M_{\rm cdm}$, $M_{\rm gas}$, T	6	✓	✓	✓	✓	LH, CV, 1P, EX

Table 3. Comparison of multifield and selected single-field cosmological generative models. n_θ: number of conditioning parameters. SI: stochastic interpolant. †: field-to-field translation conditioned on an input field rather than direct simulation-parameter conditioning; Horowitz et al. (2025) also conditions on simulation parameters in addition to the dark matter input field. Ticks indicate the metric was quantitatively evaluated; '—' indicates the metric was not reported. The absence of a metric does not imply failure; it reflects the scope and focus of each study. Andrianomena et al. (2024) reports cross-power R^2 which tests spectral shape agreement but does not isolate per-k inter-field coherence.

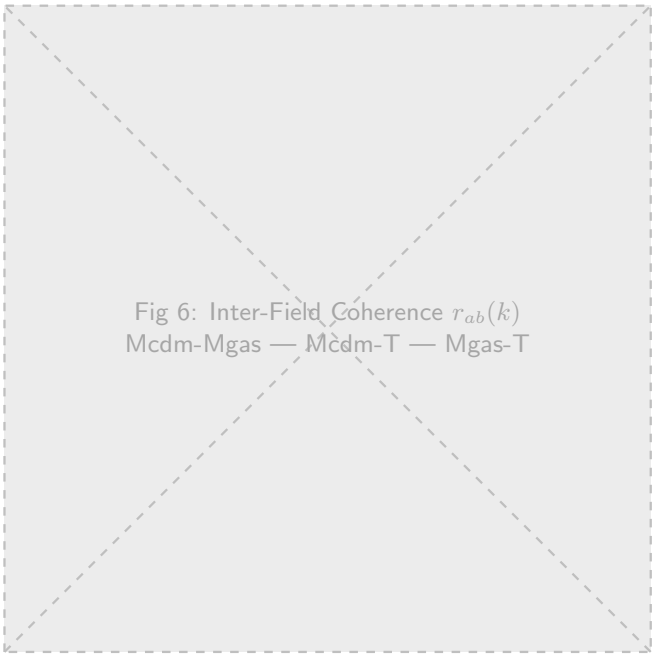

Figure 6. Inter-field coherence $r_{ab}(k)$ for generated (red) and true (black) maps at fiducial cosmology. Shaded bands show $\pm 1\sigma$ from the CV ensemble. The generated model reproduces the scale-dependent decorrelation between fields induced by baryonic feedback.

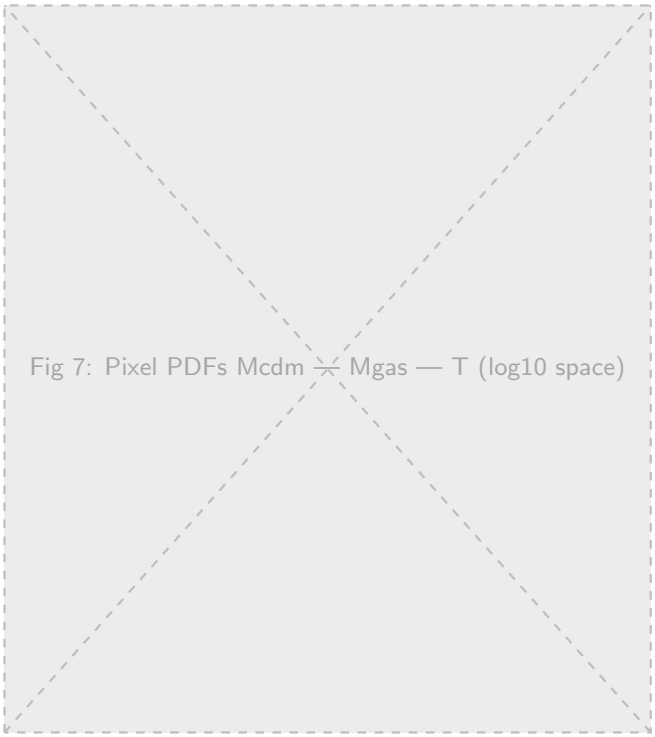

Figure 7. Pixel probability distribution functions in $\log_{10}$ space for all three fields. Black histograms: true CAMELS maps (LH test set). Red histograms: GENESIS outputs. KS statistics are reported in each panel. For T, both the cold ($\log_{10} T \approx 4$) and hot ($\log_{10} T \approx$ **[6.5]**) gas populations are reproduced.

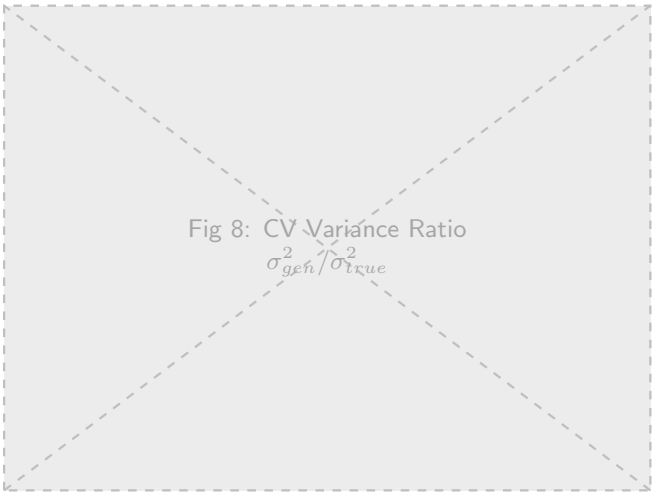

Figure 8. Variance ratio $\sigma^2_{\rm gen}(k)/\sigma^2_{\rm true}(k)$ at the fiducial cosmology for all three fields. The green shaded band marks the acceptance criterion $[0.7, 1.3]$. All three fields satisfy the > 80 per cent criterion (PASS).

CAMELS Multifield Dataset. By coupling OT-FM with a U-Net architecture designed for the periodic geometry and multifield structure of cosmological simulations, GENESIS reproduces both single-field and cross-field statistics across four validation protocols (LH, CV, 1P, EX).

Three aspects of the results are worth noting. First, dark matter auto-power residuals approach the cosmic variance floor of the $25\,h^{-1}$ Mpc CAMELS volume, suggesting that residual error is driven by training-data statistics rather than model capacity. Second, cross-power and coherence metrics remain stable across the full 6D parameter space, including regions where astrophysical feedback strongly modulates the baryonic fields. Third, the evaluation protocol — calibrated against the cosmic variance floor and spanning in-distribution, sensitivity, and out-of-distribution tests — provides a reproducible framework applicable to future multifield generative models.

These results suggest that OT-FM-based multifield emulators can serve as a useful foundation for fast cosmological forward modelling and, in the longer term, for simulation-based inference pipelines that require conditional sampling across high-dimensional parameter spaces.

We release all model weights, training code, and the evaluation pipeline at `FILL_REPO_URL`.

ACKNOWLEDGEMENTS

M.P. acknowledges support from **[funding source]**. The CAMELS simulations were run on the **[computing facility]**. This work made use of NUMPY (Harris et al. 2020), SCIPY (Virtanen et al. 2020), PYTORCH (Paszke et al. 2019), PYLIANS3 (Villaescusa-Navarro 2018), and MATPLOTLIB (Hunter 2007).

DATA AVAILABILITY

The CAMELS simulation data used in this work are publicly available at `https://camels.readthedocs.io`. The trained GENESIS model weights, training code, evaluation pipeline, and scripts to reproduce all figures in this paper will be made available at `FILL_REPO_URL` upon publication.

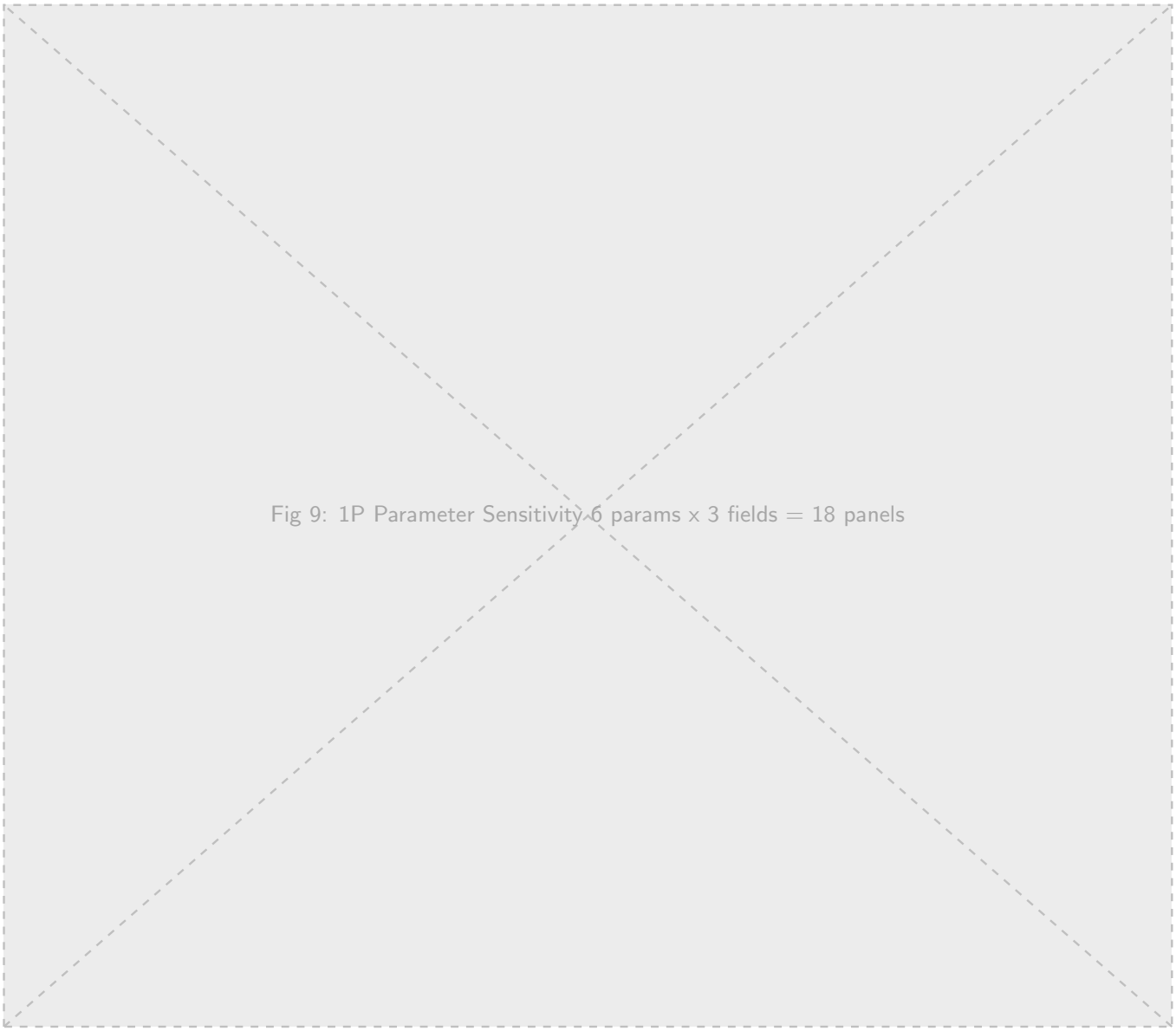

Figure 9. Parameter sensitivity curves from the 1P protocol. Each panel shows $P(k;\theta_i^+)/P(k;\theta_i^-)$ for generated (coloured lines) and true (black dashed) fields as each parameter θ_i is varied individually. Rows correspond to the six parameters; columns to the three fields. Generated curves lie within $\pm 2\sigma_{\rm CV}$ of the true curves in all panels.

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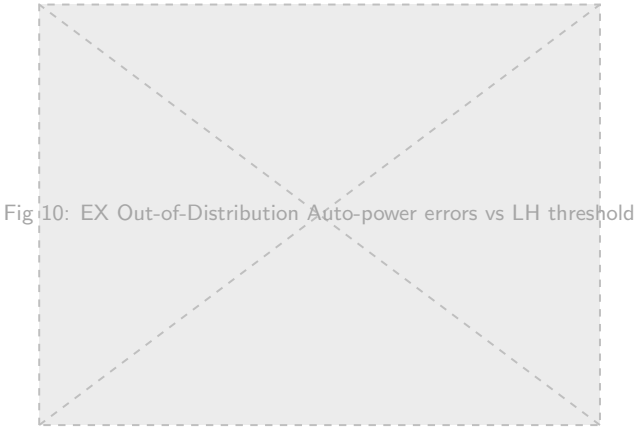

Figure 10. Out-of-distribution auto-power errors (EX protocol). Red lines: mean $\Delta P/P_a(k)$ for EX-set simulations. Grey lines: LH test-set mean for reference. Dashed and dot-dashed horizontal lines mark the LH and $2\times$ LH acceptance thresholds, respectively. No field exceeds the $2\times$ threshold (PASS).

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Channel	Transform	μ_a	σ_a
$M_{\rm cdm}$	softclip + $\log_{10}$ + affine	[X.XX]	[X.XX]
$M_{\rm gas}$	$\log_{10}$ + affine	[X.XX]	[X.XX]
T	$\log_{10}$ + affine	[X.XX]	[X.XX]

Table A1. Normalization parameters per channel. The softclip transformation for $M_{\rm cdm}$ applies a smooth saturation at the [99.X]th percentile of the training distribution before the $\log_{10}$ transform. All statistics are computed from the [850] training simulations.

Metric	U-Net	Swin-B	Swin-L	Pass thr.
$\overline{\Delta P/P}_{M_{\rm cdm}}$ (per cent)	[X.X]	[X.X]	[X.X]	<25
$\overline{\Delta P/P}_{M_{\rm gas}}$ (per cent)	[X.X]	[X.X]	[X.X]	<30
$\overline{\Delta P/P}_{T}$ (per cent)	[X.X]	[X.X]	[X.X]	<35
$\overline{\Delta P/P}_{M_{\rm cdm}\text{-}M_{\rm gas}}$ (per cent)	[X.X]	[X.X]	[X.X]	<30
$\overline{\Delta P/P}_{M_{\rm cdm}\text{-}T}$ (per cent)	[X.X]	[X.X]	[X.X]	<60
$\overline{\Delta P/P}_{M_{\rm gas}\text{-}T}$ (per cent)	[X.X]	[X.X]	[X.X]	<60
$\Delta r_{M_{\rm cdm}\text{-}M_{\rm gas}}$	[0.0XX]	[0.0XX]	[0.0XX]	<0.10
Inference time (s sample^{-1})	[X.X]	[X.X]	[X.X]	—
Parameters (M)	[XX]	[XX]	[XX]	—

Table B1. Comparison of U-Net and Swin Transformer backbones. All metrics evaluated on the LH test set. Inference time is for 25 Heun steps on a single [GPU type].

APPENDIX A: NORMALIZATION SCHEME

Table A1 summarizes the normalization parameters for each channel. The affine parameters μ_a and σ_a are computed from the training set.

APPENDIX B: SWIN TRANSFORMER COMPARISON

We compare the primary U-Net architecture against a Swin Transformer (Liu et al. 2021) backbone of comparable parameter count. The Swin variant uses the same OT flow-matching objective, conditioning scheme, and training procedure as the U-Net, differing only in the backbone (shifted-window self-attention replacing the convolutional encoder–decoder).

Table B1 summarizes the key evaluation metrics for both architectures. The U-Net achieves lower cross-power residuals for the feedback-sensitive pairs ($M_{\rm cdm}$–T and $M_{\rm gas}$–T), consistent with the hypothesis that circular convolution provides a more natural inductive bias for periodic-boundary fields. The Swin variant shows higher inference latency due to the cost of shifted-window attention at multiple scales.

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